Pharmacologic Management of Gout: Objectives

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1. Describe the mechanism of action, key drug interactions, and key toxicities of medications used for gout.
2. Develop an appropriate pharmacologic plan to treat acute gout.
3. Formulate a rational drug regimen for gout prophylaxis.

Pharmacologic Management of Gout: Objectives

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- Acute treatment: Target inflammation
  - NSAIDS
  - Corticosteroids
  - Colchicine
  - Other agents: Uricase, ACTH, IL-1 inhibitors
- Transition to ULT = Anti-inflammatory drug + ULT
- Urate-lowering Therapy (ULT): Reduce serum uric acid
  - Xanthine oxidase inhibitors – Allopurinol, febuxostat
  - Uricosuric drugs – Probenecid

Arthritis Care & Research 2020;72(6): 744-760

Pharmacologic Management of Gout: Acute Treatment

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- General principles
  - Treat with pharmacotherapy.
  - Initiate treatment within 24 hours.
  - Ongoing ULT should not be interrupted during an acute episode.

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Pharmacologic Management of Gout: Acute Treatment - NSAIDS

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- Mechanism: Inhibit prostaglandin synthesis and urate crystal phagocytosis
- Clinical use: Acute management of gout, transitioning to ULT
- Toxicities: Dyspepsia, GI ulcers, renal toxicity, bleeding
- Naproxen, indomethacin and sulindac are FDA-approved, others are used.
- Full anti-inflammatory doses (vs analgesic doses) should be used.
- **May be poorly tolerated in the elderly, renal insufficiency**

Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 36 &
Harrison's Principles of Internal Medicine, 21e, 2022; Chapters 372.

Pharmacologic Management of Gout: Acute Treatment - Corticosteroids

- <u>Mechanism</u>: Suppression of immune response
- <u>Clinical use</u>: Acute management of gout
- <u>Toxicities</u>: Insomnia, mood changes, hyperglycemia, fluid retention
- <u>Dose</u>:
  - Oral: recommended if 1 or 2 joints involved
    - Prednisone: 0.5 mg/kg/day for 5-10 days at full dose ***OR*** 2-5 days at full dose then taper for 7-10 days
  - Intra-articular (IA) if 1 or 2 large joints
    - Can be used in combination with oral steroid, colchicine or NSAIDS
    - IA steroids may appear as crystal in joint aspirate
  - Intramuscular (IM)

Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 39 & Arthritis Care & Research. 2020; 72(6): 744-760.

Pharmacologic Management of Gout: Acute Treatment - Colchicine

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- <u>Mechanism</u>: Inhibits inflammatory response by inhibiting microtubule polymerization & inflammatory mediators
- <u>Clinical use</u>: Acute management of gout, transitioning to ULT; may be used for long-term prophylaxis
- <u>Toxicities</u>: Nausea, vomiting, diarrhea, (neuropathy, bone marrow suppression)
- Avoid in patients with renal or hepatic impairment <u>who are taking P-gp or strong CYP3A4 inhibitors.</u>
    - In these patients life-threatening and fatal colchicine toxicity has been reported with colchicine taken in therapeutic doses.

Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 36 & Colcrys® Product Information.

Pharmacologic Management of Gout: Acute Treatment - Colchicine

- Recommended if initiated within **36 hours** of symptom onset
- Dose: 1.2 mg (two 0.6 mg tablets) once, then 0.6 mg in 1 hour
  - Total dose is three 0.6 mg tablets. Please know this dose.
  - This dose is lower than that historically recommended but is equally efficacious and less toxic.

Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 36.

Pharmacologic Management of Gout: Prophylaxis (Urate-Lowering Therapy)

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- ULT initiation is recommended if:
    - ≥ 1 tophi
    - Evidence of radiologic damage
    - Frequent attacks (≥ 2 attacks/year)
- <u>ULT may be initiated during an acute attack if an anti-inflammatory medication has been initiated; ULT alone may induce gout flare</u>
- Dose ULT therapy to target a serum uric acid <6 mg/dL.

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Pharmacologic Management of Gout: Prophylaxis – Xanthine Oxidase Inhibitors (XOIs)

- Although most patients under-excrete uric acid (vs over-produce), xanthine oxidase inhibitors are commonly prescribed due to ease of use and efficacy.

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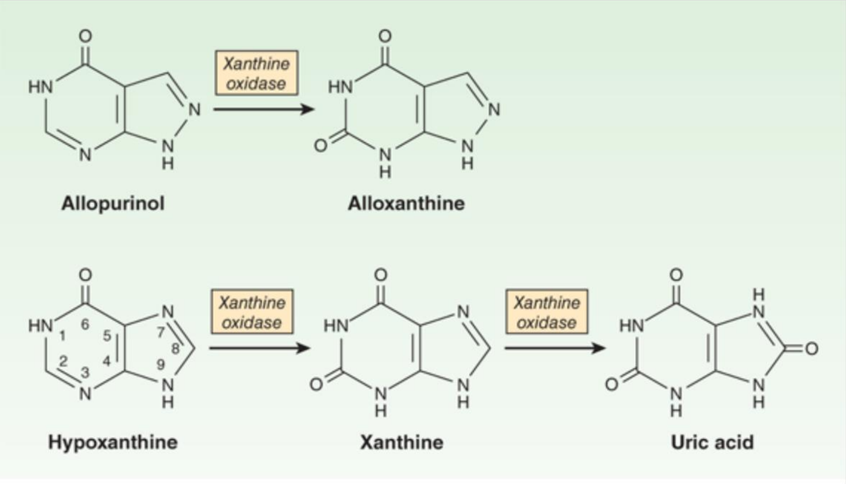

Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 36 (Fig 36-7), Arthritis Care & Research 2020;72(6): 744-760

Pharmacologic Management of Gout: Prophylaxis (XOI) - Allopurinol

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- Mechanism: Reduces uric acid formation through competitive inhibition of xanthine oxidase
- Clinical use: Chronic management of gout (typically 1st line); also indicated for cancer treatment-related hyperuricemia
- Metabolism: Metabolized by xanthine oxidase to oxypurinol (alloxanthine), which also inhibits xanthine oxidase

Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 36

Pharmacologic Management of Gout: Prophylaxis (XOI) - Allopurinol

- Toxicities:
  - Most adverse effects are rare, mild and reversible: dyspepsia, headache, diarrhea
  - If rash develops patient should STOP the medication!!
  - Hypersensitivity is an infrequent complication associated with the HLA-B*5801 allele (rash, eosinophilia, leukocytosis, fever, hepatitis and progressive renal failure)
- Initial dose: 100mg/day
  - Can initiate in CKD ≥ stage 3; need to renally adjust dose
- Drug interactions: Azathioprine, mercaptopurine
- Monitoring: Liver function tests (LFTs), serum creatinine (Scr), CBC

Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 36

Pharmacologic Management of Gout: Prophylaxis (XOI) - Febuxostat

- Mechanism: Non-purine inhibitor of xanthine oxidase
- Indications: Chronic management of gout
- Metabolism: Hepatic oxidation and conjugation to inactive metabolite; no renal adjustment necessary
- Toxicities: ↑ LFTs, nausea, musculoskeletal pain, rash
- Black box warning (2019): Increased risk of CV death in patients with history of CVD

Uloric® Product Information & *NEJM* 2018;378:1200-10.

Pharmacologic Management of Gout: Prophylaxis (Uricosuric) - Probenecid

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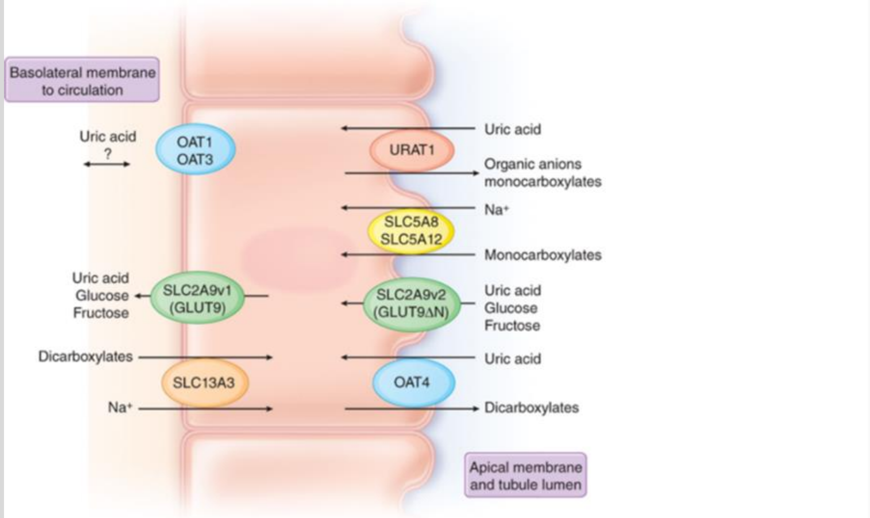
- Uricosurics compete with uric acid for the transporter, inhibiting reabsorption in proximal tubule
- Urine volume must be maintained
- Avoid in:
  - Uric acid over-producers
  - Urolithiasis



Source: D. L. Kasper, A. S. Fauci, S. L. Hauser, D. L. Longo, J. L. Jameson, J. Loscalzo: Harrison's Principles of Internal Medicine, 19th Edition.
www.accessmedicine.com

Harrison's Principles of Internal Medicine, 21e, 2022; Chapter 372, Figure 417-2

Pharmacologic Management of Gout: Prophylaxis (Uricosuric) - Probenecid

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- Mechanism: Competitively inhibits URAT1
- Indication: Chronic management of gout
- Toxicities: Rash, precipitation of gouty arthritis, nephrolithiasis
- Drug interactions: Reduces excretion of weak acids (penicillins)
- Not effective if eGFR <60 mL/min

Katzung & Trevor's Basic and Clinical Pharmacology, 16e, 2024; Chapter 36 &
Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 42